

An all-round AI-Chemist with scientific mind

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ABSTRACT

The realization of automated chemical experiments by robots unveiled the prelude of artificial intelligent laboratory. Several AI-based systems or robots with specific chemical skills have been demonstrated, but conducting all-round scientific research remains challenging. Here, we present an all-round AI-Chemist equipped with scientific mind that is capable of performing all essential steps required for chemical research. Based on a service platform, the AI-Chemist is able to automatically read the literatures from cloud database and propose experimental plans accordingly. It can control a mobile robot in house or online to automatically execute the complete experimental process on 14 workstations, including synthesis, characterization, and performance test. The experimental data can be simultaneously analyzed by the computational brain of the AI-Chemist through machine learning and Bayesian optimization, allowing to propose new hypothesis for next iteration. The competence of the AI-Chemist has been scrutinized by three different chemical tasks. All-round AI-Chemists with scientific mind may dramatically change the landscape of the chemical laboratory in the future.

Keywords: AI-Chemist, computational brain, mobile robot, machine reading, all-round research

INTRODUCTION

Empowering a robot with scientific mind to help humans decipher the high-dimensional correlations of the complex world, reduce the Try-Error development cost of new materials, and even realize interstellar exploration and colonization has been a fantasy in many science fictions [1, 2]. The recent advancement in automated chemical experiments by robots offered glimpse of the hope [3-15]. An automated chemical machine system (called Chemputer) has been developed to integrate literature reading, protocol customization, organic synthesis and characterization [9, 10], while another more generalized system could carry on lego-like automated organic synthesis [11, 12]. In 2020, a cloud-native chemical platform with artificial intelligence as its controller was reported for organic retrosynthesis based on previously available data [13, 14]. Meanwhile, the research group led by Cooper has pushed a step further to develop a mobile robotic chemist with the abilities to perform experiments faster than human and to select photocatalysts with Bayesian optimization [15]. But, they also specifically mentioned in their article that the robotic search employed in their work does not capture existing chemical knowledge, nor include theory or physical models. A better approach as they suggested is to equip with computational brain and to fuse theory and physical models with autonomous searches [15]. Indeed, only after inserting a computational brain, the robot can truly be self-controlled by its scientific mind to conduct all-round scientific research.

Along this line, we have built an all-round AI-chemistry laboratory that can (1) propose scientific hypothesis and generate experiment plans using existing knowledge, (2) execute the complete experimental procedure (synthesis, characterization, and performance testing) for multiple chemical tasks, (3) build predictive models exploiting theoretical computations and experimental data feedback, so as to execute all-round chemical research with intelligence. Fig. 1A outlines the overall architecture of the all-round AI-Chemist. This AI-Chemist with human-like research wisdom consist of three modules, including machine reading module to capture existing chemical knowledge by automatically reading massive chemical literatures, a mobile robot module to produce experimental data by executing various chemical experiments, and a computational brain module to generate physics/theory-based predictive models by carrying out theoretical calculations. The completion of the three modules leads to the birth of an all-round AI-Chemist with scientific mind.

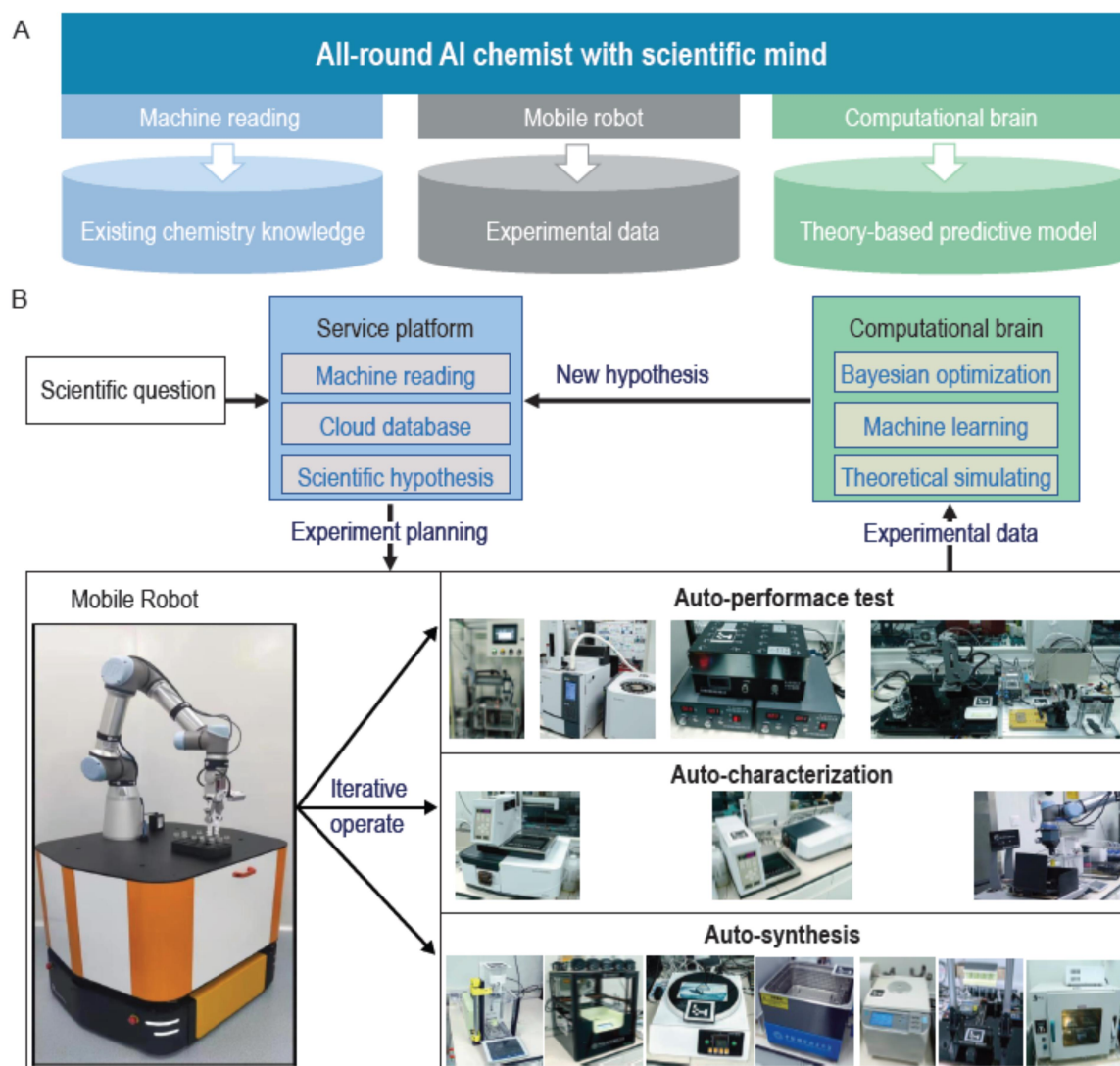


Figure 1. Design of all-round AI-Chemist with scientific mind. (A) Three modules of AI-Chemist: a machine reading module, a mobile robot module, and a computational brain module. (B) The workflow of AI-Chemist and the functions of each module.

RESULTS AND DISCUSSION

The workflow of this all-round AI-Chemist forms a complete closed loop (Fig. 1B), and is controlled by a home-developed system software (Fig. S1). When the AI-Chemist is prompted with a scientific question raised by a human researcher, its service platform with user-friendly graphical user interface (GUI) would propose a scientific hypothesis after capturing existing knowledge by machine reading of massive literatures. This service platform possesses the ability of accessing the cloud databases, designing experiment plans and the web-based feature, which enable to easily manage tasks and remotely monitor the mobile robot. The experiment plans generated by this service platform are provided to the mobile robot and various smart chemical workstations. A control system software based on Robot Operating System (ROS) [16] is developed to control the mobile robot with a six degree of freedom robotic arm to move around and operate all apparatuses. The laboratory hosting the robot and apparatus is divided into an auto-synthesis area, an

auto-characterization area, and an auto-performance-testing area, respectively. All instruments were designed to operate using vials of the same type, so as realizing the combination of multiple apparatuses to form a complete workstation. Many previous synthesis robots mostly optimize experimental plans according to previous data in literatures [9] or the data generated by robot itself [15], while the AI-chemist can “think” more deeply. Thanks to the computational brain it possesses, it can not only capture existing chemical knowledge and accept experimental feedback, but also preform theoretical simulations, train machine learning models, and carry out Bayesian optimizations to provide global optimal solution. This robot can propose a new hypothesis and new experiment plan by itself, as well as carry out next round of chemical experiments by itself.



displayed on the service platform. (E) Experimental archives for building experimental database and training prediction model.

The first task of an AI-Chemist is to read through many literatures to gain the wisdom of human chemists. By digitizing and standardizing experimental protocols written in natural language from the literatures, the existing knowledge can be transferred to the robot to enrich its brain. As shown in Fig. 2A, based on syntax rules, domain dictionaries, and machine learning, we build a scientific text data mining system with eight natural language processing (NLP) steps. This system can decompose titles, paragraphs, and sentences by text classification, locate important regions by word tokenization and part of speech tagging, name entity recognition, perform grammatical analysis, extract entity relationships, and realize coreference resolutions and error corrections. The NLP steps convert scientific texts into structured data understandable for AI. This data mining system belongs to a service platform with web browser-based human machine interface (HMI), which can further propose hypothesis and provide intelligent recommendation of experiment plans according to above structured data. A user-friendly GUI helps researchers remotely monitor the status of robot and workstations (Fig. 2B and Fig. S2), and customize the experimental procedure by clicking and dragging on the GUI. As shown in Fig. 2C, according to the scientific hypothesis, the service platform communicates with all workstations to check their status and customizes experiment workflow. We have also generated a cloud-based chemical database to facilitate choosing chemical samples from a solid or liquid dispensing menu with a dragging down list (Fig. S3A). The database can also be accessed through a web interface in searching for any compound, such as terephthalic acid shown in Fig. S3B. Once the experiment workflows are validated, they are stored as experimental templates to cloud database for subsequent experiment recall. The service platform can also carry out preliminary analysis of experimental data collected, and the results of characterization and performance testing are displayed visualized as graphs whenever possible on the browser-based interface (Fig. 2D and Fig. S4-S9). These experimental data are also used to establish and update a database for subsequent prediction models and Bayesian optimizations (Fig. 2E). The AI-Chemist is able to automatically iterate experimental conditions until experimental results reach threshold parameters set previously (Fig. S10).

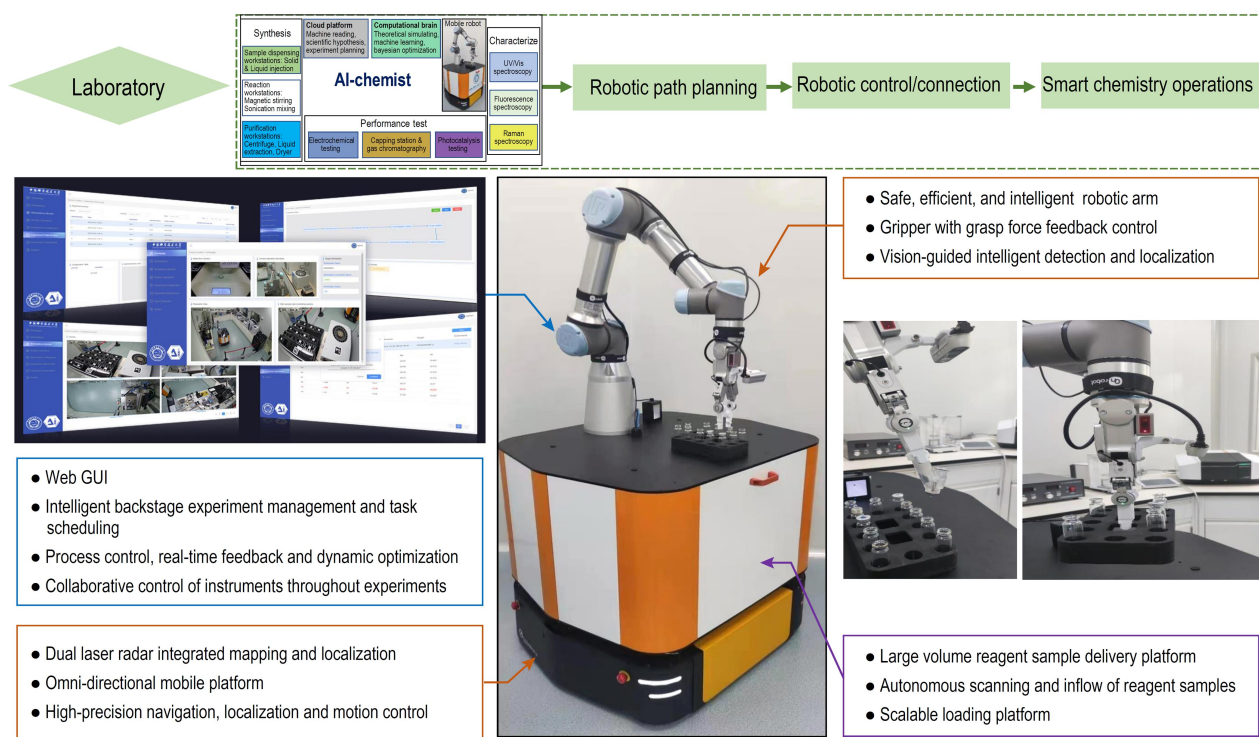


Figure 3. Intelligent chemical laboratory and mobile robot. The intelligent chemical laboratory with capabilities of robotic path planning, robotic control and connection, and smart chemical operations and the mobile robot equipped with control system, dual-lidar-based localization system, six degree of freedom robotic arm, and large scalable loading platform.

To execute the whole process of synthesis, characterization and test, we have set up a mobile robot and 14 workstations in the laboratory, including auto-synthesis region, auto-characterization region, and auto-performance-test region, respectively. When AI-Chemist executes a synthesis task, it uses the liquid dispensing workstation with accuracy of 3 μL and solid dispensing workstation with accuracy of 0.1 mg to prepare the reagents (Fig. S11-S12). It then uses magnetic stirring workstation and sonication mixing workstation with time control accuracy of millisecond to make the desired reactions (Fig. S13-S14). Finally, the purification of products is conducted at dryer workstation, centrifugation workstation and liquid extraction workstation (Fig. S15-S17). After completing the synthesis process, the sample vessel with products is transferred to auto-characterization region and auto-performance-test region. Currently, the laboratory is equipped with auto-controlled UV-Vis, fluorescence, and Raman spectroscopy workstations to characterize the component and structure of products (Fig. S18-S20), as well as auto-controlled electrochemical workstation, capping workstation, photocatalysis workstation, HS gas chromatography (GC) to test the catalytic performance of products (Fig. S21-S24). A home-developed system software capable of robotic path planning, robotic control and connection, and smart chemical operations is in charge of coordinating the real-time interactions between the robot and the workstation. The mobile robot simply executes all orders in sequence, and each workstation only needs to ensure an experimental operation is executed using accurate parameters and the experimental data are feedback to the system correctly. Together, the robot and the workstations formed a holistic smart laboratory for the AI-Chemist (Fig. 3).

To ensure precise and interactive chemical operations, the mobile robot is equipped with a

dual-lidar-based integrated mapping and localization system to obtain its localization information and laboratory map for the navigation and autonomous obstacle avoidance (Fig. 3 and Fig. S25). An omni-directional mobile platform with global localization precision of 10 mm and max loading of 200 kg ensures the motion accuracy and load ability of the robot. A large and scalable loading and delivery platform has been ready for high-throughput and multi-tasking experiments. The chemical operations are performed using a six degree of freedom robotic arm (Fig. 3). It is equipped with gripper having grasp force feedback control, depth cameras and laser sensors, and vision-guided intelligent detection and localization system. To fulfill the requirements of high localization accuracy of the gripper in experimental operations, an independent ArUco label is attached on each experimental workstation (Fig. S26-S30). The robot can recognize the current workstation by visually identifying the label ID in the field of view, and make the accurate measurement of the workstation pose according to the relative pose between the ArUco label and the target to be operated on the experimental workstation. To handle abnormal situations during the experimental operation, such as detecting and locating transparent vials in unstructured scenarios, a set of algorithms based on deep learning has been developed (Fig. S31-S32). A task management module is of central importance in the whole robot control system. It is responsible to detect the current status of robot and workstations and plan experimental operations (Fig. S33-S34). A video presents the overview of this intelligent, full-process chemical laboratory (Supplementary Video 1). Compared with existing literature reports on automated synthesis systems, characterization and related performance testing, our AI-Chemist incorporates computational brain and chemical workstations, and is therefore able to achieve more chemistry functions and higher control precision (Table S1).

Two experiments were performed to examine the hardware and software of robot and workstations. In the first experiment, the AI chemist is instructed to find a biocompatible luminophore with aggregation-induced emission (AIE) characteristics [17-19]. The machine reading module is motivated: from 15,979 papers, 4865 molecules are found to relate with "AIE", "aggregation-induced emission", "aggregation induced emission". After data cleansing based on expertise rules (See details in Materials and Methods), 306 molecules are identified to be candidates as the commercially available AIE luminophores. Fig. 2A and Table S2 lists the top 15 molecules with high occurrence frequencies (See Supplementary Data 1 for complete list). Among them, Berberine chloride (BBR) [20, 21] is the only one with emission wavelength located in the visible region, and therefore chosen for further investigation. A series of automated experiments are conducted by the AI-Chemist, to synthesis different berberine chloride solutions and measure their fluorescence. The robot weights an appropriate amount of solid berberine chloride sample using the solid dispensing workstation, and then transfers it to the liquid dispensing workstation to dissolve it into a solution. The solutions' optical properties are measured with Photoluminescence (PL) spectroscopy and UV spectroscopy. As shown in Fig. 4A-C, BBR solutions of different concentrations and solvents are compared, and the optimal concentration of BBR is thus identified to be 20 mM.

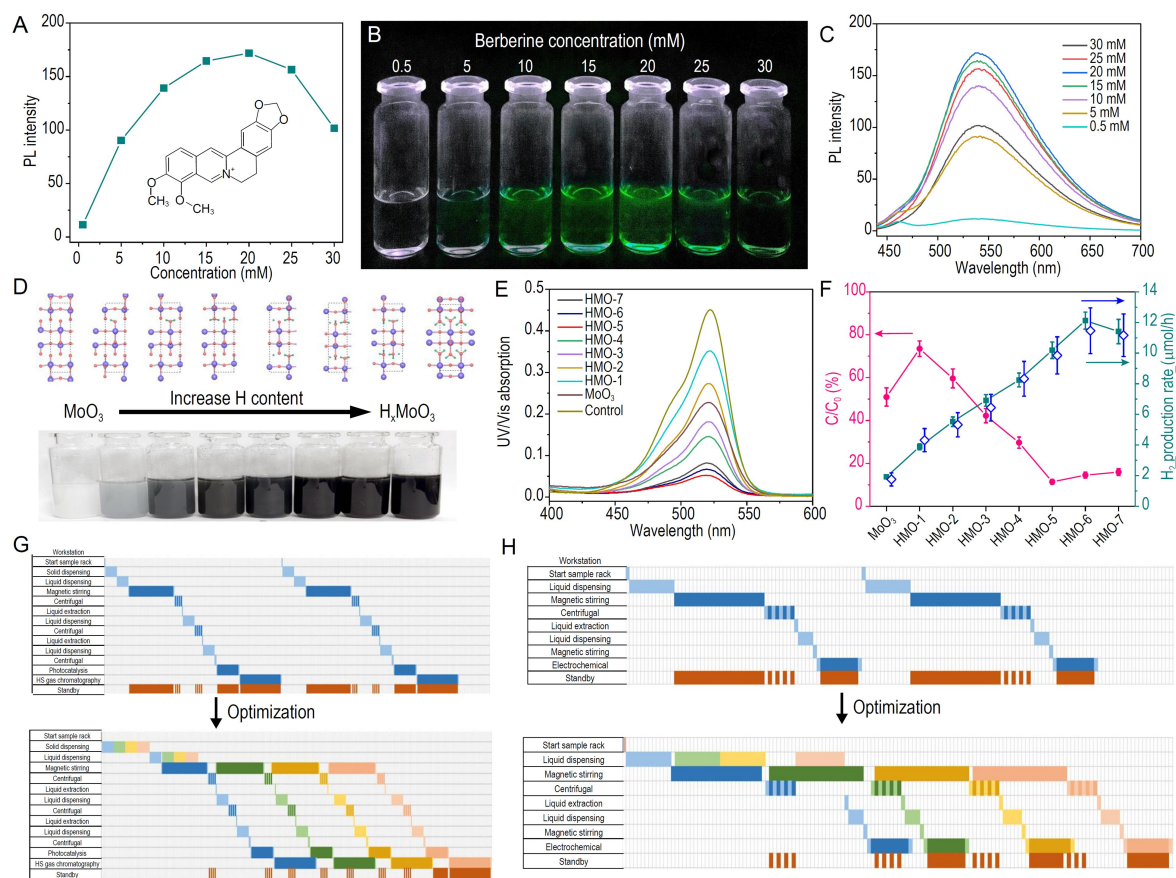


Figure 4. Chemical experiments performed by mobile robot and workstations. (A) Peak fluorescence intensity with increased concentrations of berberine chloride (BBR); (B) Fluorescence images of the solution in (A) under UV light (365nm); (C) PL spectra of BBR chloride in solution with different concentrations of BBR chloride. Excitation wavelength: 405nm. (D) Tunable hydrogenation of MoO_3 by Cu-acid treatment with simulated crystal structure of H_xMoO_3 with different concentration of H-dopants in the lattice. (E) The UV-Vis spectra of RhB solution after photocatalytic degradation. (F) Characterization of photocatalytic degradation efficiency of RhB (left column, reaction conditions: $[\text{cat}] = 0.1 \text{ g}$, solution volume: 10 mL H_2O , $[\text{RhB}] = 10^{-5} \text{ mol/L}$) and the yield of dye-sensitized photocatalytic H_2 production (right column, reaction conditions: $[\text{Eosin-Y}] = 0.6 \text{ mM}$, $[\text{TEOA}] = 1.5 \text{ M}$, $[\text{cat}] = 0.1 \text{ g}$, solution volume: 10 mL. All the light sources are 25W white LED lamps). The error bars show the systematic average error and the comparison with the manual work result (blue error bar) (G) Schematic diagrams of experiment flow for dye-sensitized photocatalytic water splitting experiment without (top) and with (bottom) the multi-task dynamic optimization. (H) Schematic diagrams of experiment flow for electrocatalytic oxygen evolution reaction experiments without (top) and with (bottom) the multi-task dynamic optimization.

The second experiment is to optimize hydrogen doping strategy of metal oxide photocatalysts. A set of H_xMoO_3 samples with various amount of hydrogenation via Cu-acid treatment [22] are produced by the synthesis workstations (Fig. 4D), and then used for photocatalytic degradation of rhodamine B (RhB) at the photocatalysis workstation. The photocatalytic reactions are monitored by the UV-Vis spectroscopy workstation (Fig. 4E): the photocatalytic degradation efficiency of RhB reaches to a maximum level on the

sample HMO-5. To obtain the yield of dye-sensitized photocatalytic H_2 production during the reaction, the vials are vacuum sealed at the capping workstation and transferred to HS gas chromatography (GC) workstation for measurement of H_2 produced. The sample HMO-6 shows the highest H_2 production rate, as shown in Fig. 4F. Here the average of deviation bars for all experimentally measured points are 5.5% for H_2 production rate and 8.3% for RhB degradation efficiency (Table S3), suggesting highly accuracy and repeatability of the AI-Chemist. Based on the task management module, a smart chemical system capable of multi-task dynamic optimization is developed. With its help, the total experiment time and waiting time of the robot can be significantly shortened: the total time consumption of the 4 sample racks of dye-sensitized photocatalytic water splitting experiments without optimization is 1810 minutes, while the total time consumption of the 4 sample racks of experiments after multi-task dynamic optimization is 980 minutes (Fig. 4G and Supplementary Text).

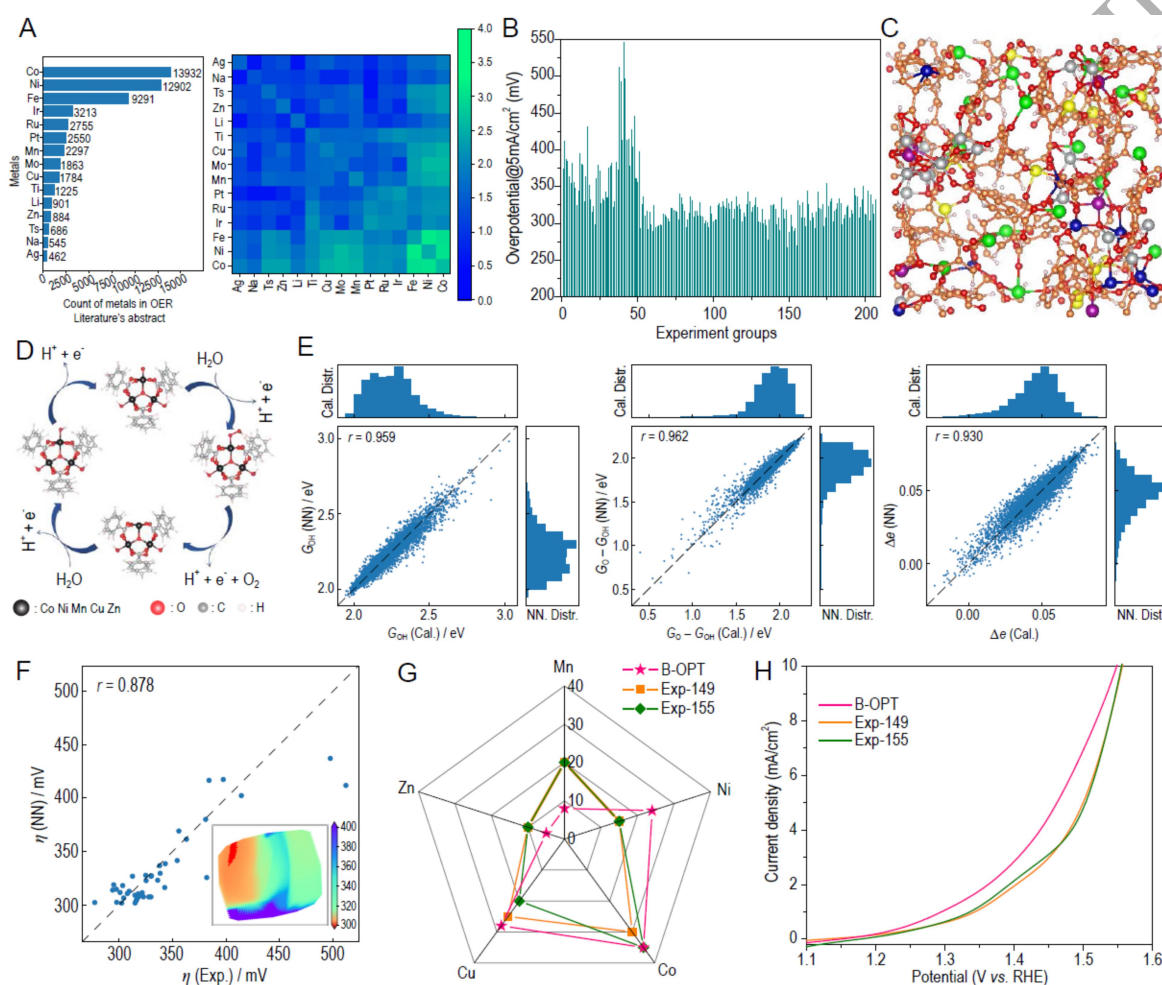


Figure 5. An all-round chemical research conducted by AI-Chemist. (A) The order of metal recommendation and the frequency of metal co-occurrence by machine reading. (B) The overpotential values of 207 Try-Error experiments carried out by mobile robot and workstations. (C) An example of simulated structure generated by molecular dynamics. (D) The simulated OER reaction path. (E) The prediction results of three catalytic properties by neural network models, where G_{OH} is free energy change of hydroxyl adsorption, G_{O} is free energy change of oxygen atom adsorption, Δe is the charge transfer during hydroxyl adsorption. (F) The prediction results of overpotential by neural network model calibrated by experimental data. Inset:

dimensionality reduction plot by principal component analysis (PCA) for predicted overpotentials of all exhaustive samples. (G) Kiviat diagram of composition ratios and (H) polarization curves of the optimal sample suggested by Bayesian model and the best samples by Try-Error experiments.

The above-mentioned two tasks have proven that the mobile robot and smart workstations can successfully execute various experimental operations. Next, we integrate them with the service platform and computational brain to act as an AI-Chemist to conduct all-round chemical research. We take the non-noble metal oxygen evolution reaction (OER) electrocatalyst as an example. It has been shown that high-entropy alloy nanoparticles containing more than four different elements have the high potential to be the choice of future electrocatalysts due to their structural stability, diversity of adsorption sites, and remarkable catalytic activity [23-26]. Among them, the MIL-101 MOF is an ideal platform for different metal elements to synthesize high-entropy electrocatalysts [27]. However, the wide range of choices in element combinations poses grand challenges for developing high-entropy electrocatalysts. Even if we only want to determine a desirable metal combination, it could take hundreds of thousands of years to traverse the best composition ratio. This simple fact makes high-entropy alloy nanoparticles a suitable task for our all-round AI-Chemist with scientific mind. Again, the service platform sorts out metal recommendations via intelligent machine reading around 16,000 papers from the cloud database (Fig. 5A and Supplementary Data 2). Five non-precious metal elements are selected to construct 207 Try-Error experiments fully executed by the mobile robot on the smart chemical workstations. The home-developed smart chemical system successfully shortens the total time consumption of the four groups of experiments from 630 minutes to 365 minutes through multi-task dynamic optimization (Fig. 4H and Supplementary Text). The experimental results summarized in Fig. 5B show that the No. 149 and No. 155 samples have the best electrocatalytic performance (Fig. S36-S37). We directly carry out Bayesian optimizations based on the experimental data to determine the optimal composition ratios. Two optimized solutions are suggested (Fig. S38-S39), whose composition ratios and performances are quite close to the best results of the above-mentioned experiments (Fig. S40). This could imply that these are only local optimal solutions due to the blindness of Bayesian optimization based on the experimental data [15]. To take advantage of computational brain inserted in our AI-Chemist, we have extended our search based on theoretical calculations. Herein, around 20,000 molecular dynamics simulations are performed by the computational brain module to generate a large number of structures (Fig. 5C and Fig. S41) and a high-quality simulation dataset through density functional theory calculations (see details in Materials and Methods, Table S4). According to the structure around the metal node in the MIL-101, we calculate the overpotentials based on the reaction path shown in Fig. 5D and Fig. S42. The trend of simulated overpotential values is consistent with the experiment, but there are systematic errors for the absolute value, which needs to be corrected by combining simulations and experiments (Fig. S43). Eventually, a machine learning model for predicting the key electrocatalytic properties from metal composition ratios is established by using the simulated data (Fig. 5E) and is calibrated with the Try-Error experimental data to obtain an accurate prediction model for real experimental overpotentials (Fig. 5F). This calibrated model can rapidly and massively predict the composition-ratio-based overpotentials (inset of Fig. 5F) that can be used to evaluate the electrocatalytic performance of high-entropy materials. Again, the Bayesian optimization algorithm is applied to narrow down the huge searching space (553,401 options) and find out the optimal solution. From the Kiviat Diagram, one can see the optimal composition ratio suggested by Bayesian model is far from the best samples of above-mentioned experiments (Fig. 5G), which indicates that the Bayesian optimization based on the combination of simulated and experimental datasets can go beyond the local optimal regions. The sample with the best composition ratio is then synthesized and characterized by the AI-Chemist, and its overpotential is

231.7 mV at current density of 5 mA/cm², which is considerably better than the results of previous Try-Error experiments (Fig. 5H and Fig. S44). We also notice that the composition ratio suggested by Bayesian optimization is almost the same as that suggested by grid-point scanning (Fig. S45), proving that it is indeed a global optimal solution. This task has serviced as a very good example to illustrate the superior ability of the all-round AI-Chemist with scientific mind. As presented in Supplementary Video 2, the scientific mind of AI-Chemist comes from its computational brain, which is able to calculate the physical parameters based on physical models such as MD and DFT. The computational brain can further establish machine learning models based on big data and perform global search based on Bayesian optimization. From millions of possibilities, the new scientific hypothesis for the optimal solution was proposed and verified experimentally. We also manually synthesized the optimal material and found its OER performance is basically consistent with the results made by AI-Chemist (Fig. S46), confirming that the experimental results of our intelligent system are highly stable, reliable, trustworthy and reproducible.

CONCLUSION

The all-round AI-Chemist with scientific mind reported here consists of service platform, mobile robot, workstations, and computational brain. It is able to read literatures, to propose hypothesis, to design experimental plans, to execute automated operations, to analyze experimental data, to train machine learning models, and to feedback new hypothesis. In other words, it possesses all the abilities to conduct high level chemical research, which can only be possible for a group of well-organized human chemists. The closed-loop iterative design of our AI-Chemist shows its versatility in the fields of electrocatalysts, photocatalysts, and luminescent materials. The AI-Chemist possesses a universal software protocol and standardized hardware interface. This modular design makes it extendable to meet the requirements of various experimental tasks via adding more experimental workstations or computational mechanisms in the future. The fact that optimal materials could be intelligently screened out or designed by the AI-Chemist can certainly greatly reduce the time for human chemists to do experiments. It should also be mentioned that AI-Chemist can only acquire information from the existing knowledge and conduct experiments within the known techniques. The future discovery still largely depends on the human scientists to develop new theory and invent new technology. Nevertheless, it is safe to say that the AI-Chemist has begun to change our ways of finding and making new materials. The all-round AI-Chemists with scientific mind may completely change the conventional chemical laboratory in the coming years.

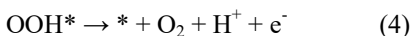
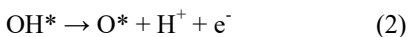
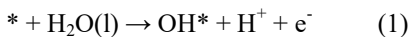
METHODS

DFT calculation

The cluster of MOF is obtained based on the crystal structure of MIL-101. Subsequently, all the DFT calculations of electrical structures are adopted by the Vienna ab-initio simulation package (VASP) [28]. Considering the long-range van der Waals (vdW) interaction corrections, the Grimme's DFT-D3 method is applied with an energy cutoff of 400 eV. The Perdew-Burke-Ernzerhof (PBE) functional [29] is applied and the projector augmented wave method (PAW) [30] is implemented as the basis set for the plane-wave expansion. The convergence of forces and energies for the geometric structures are set to 0.01 eV/Å and 10⁻⁵ eV, respectively. The Brillouin zone is sampled with 1 × 1 × 1 Monkhorst-Pack k-mesh with the vacuum space of ~15 Å.

Calculation for the free energies

Based on the general single site of oxygen evolution reaction (OER) mechanism, the four electron-proton transfer steps involved in the whole reaction could be written as:



To calculate the free energy changes, a standard change of Gibbs free energy at zero potential was calculated according to the following equation:

$$\Delta G = \Delta E + \Delta \text{ZPE} - T\Delta S$$

where ΔE refers to the change of reaction energy based on DFT simulations, ΔZPE is the zero-point energy change calculated by the vibrational frequency using the finite difference method, and ΔS represents the entropy change for each elementary step. The temperature in our work is set to be 298.15 K.

Molecular dynamics simulations

Classical molecular dynamics (CMD) simulations are performed with the LAMMPS package [31] in the isothermal-isobaric (NPT) ensemble via the Nose-Hoover barostat and thermostat [32, 33]. To accelerate the structural sampling, the CMD simulations are run at 3000 K with a timestep of 1 fs. The initial simulation box of $4 \times 4 \times 4 \text{ nm}^3$ has periodic boundaries in the all directions. 60 different metal atoms, 20 oxygen atoms, and 40 terephthalic acid ions are randomly placed into the box. The force-field parameters of them are all generated by the LAMMPS Interface program [34]. The cutoff distance for Lennard-Jones and Coulombic potential is 12.5 Å. The duration of simulation is 1 ns.

Expertise rules for AIE luminophores data cleaning

To identify the practically available AIE luminophores, several expertise rules are applied. Firstly, the CTE molecules and common solvents are removed, leaving 306 molecules. Then, two more criteria are applied to further shorten the candidate list: water solubility and frequency of fluorescence light. The top entry is tetraphenylethylene, a well-known and probably most studied organic AIE dye [35-37]. An ML model is trained to predict water solubility of all molecules based on their SMILES. After removing ones are expected to insoluble or sparsely soluble in aqueous solution, the list reduces to 20s. The fluorescence frequencies of each of them are predicted by calculating the energy gap between LUMO and HOMO using DFT. Only molecules with fluorescence located in visible range make the final list. The top entry is berberine (or berberine chloride).

Neural network architecture

The first neural network (NN) applied different metal ratios as descriptors to predict the catalytic properties (GOH^* , $\text{GO}^*\text{-OH}^*$, ΔE). To obtain the relations between catalytic properties and real overpotential, the second NN with the catalytic properties as descriptors was implemented to predict the real overpotential. One input layer, three hidden layers and one output layer constituted the whole of NN. The number of neurons in the hidden layer was 128, 128, and 128, respectively. To establish the correlation between the descriptors and prediction targets, the dataset (80% for training and 20% for validation) was trained with the Rectified Linear Unit activation function [38]. The dropout parameter [39] was used to prevent overfitting of the second NN.

The backpropagation algorithm implemented in TensorFlow [40] enabled it to track the weights successfully and update them rapidly.

SUPPLEMENTARY DATA

Supplementary data are available at NSR online

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AUTHOR CONTRIBUTIONS

W.S., J.J., and Y.L. directed the project; Q.Z., L.Z., M.L., S.W., J.L., G.Z. designed the chemistry experiment and prepare the necessary chemical compounds; F.Z., X.Z., T.S., X.T., X.L., G.H., B.C., J.Z., Y.Z., S.C., W.S. built the mobile robot system for chemistry research; Y.H., B.Z., S.W., D.Z., H.L., J.J. built the computational brain for AI-Chemist; H.X., L.Z., J.C., G.Y., W.Z., X.C., J.J. designed and constructed those smart chemistry stations; Q.Z., F.Z., H.X., M.L., G.Y., W.Z., J.J. controlled the AI-Chemist to accomplish all the chemistry tasks; Y.L., J.J., W.S., Q.Z., F.Z., Y.H., H.X. conceived the principle idea and co-wrote the paper. All authors discussed the results and commented on the manuscript.

Conflict of interest statement. None declared.

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